

X- Θ Holonomy: A Geometric, Path-Dependent Extension of Bell Correlations

Divyang Panchasara*¹

¹Independent Research / Nayvid Technology LLP, Bengaluru, India

January 4, 2026

Abstract

We introduce $X-\Theta$, a geometric framework in which Bell-type correlations arise from a hidden phase connection defined over detector configuration space. In its static formulation (Version-0A), the framework reproduces Tsirelson-saturating CHSH correlations without invoking superluminal signaling or measurement dependence. We then present a dynamic extension (Version-0B) in which the effective phase acquires bounded, path-dependent transport, leading to an operational asymmetry between clockwise (CW) and counter-clockwise (CCW) traversals of the CHSH settings loop. The resulting holonomy-induced asymmetry preserves no-signaling and local flat marginals, while providing a falsifiable geometric signature beyond standard assumptions of path-independence. We discuss experimental tests based on controlled cycling of settings and outline principled future refinements (curved memory geometry, non-Abelian transport, dwell-time dependence, and correlator-level diagnostics).

Contents

1	Introduction	1
2	Bell scenario, CHSH, and configuration space	2
2.1	Operational scenario	2
2.2	Configuration space and paths	2
3	Axioms of $X-\Theta$	2
4	Version-0A: Static connection and Tsirelson saturation	2
4.1	Static effective phase	2
4.2	CHSH and a geometric view of the plaquette	3
5	Version-0B: Dynamic holonomy via bounded memory transport	3
5.1	Bounded memory variable	3
5.2	Operational holonomy and CW/CCW asymmetry	3
6	Empirical summary from simulations	3
7	No-signaling and consistency checks	4
8	Experimental outlook	4

*Email: your-email@example.com

9	Future refinements (physics, not tuning)	4
10	Conclusion	5
A	Notes on interpretation and falsifiability	5

1 Introduction

Bell inequality violations are often interpreted as requiring nonlocal influences or contextuality. Yet modern physics contains many operational effects—Berry phase, Aharonov–Bohm interference, Wilson loops—in which *global geometric structure* produces observable signatures without local forces or superluminal signaling. This motivates a geometric alternative viewpoint: Bell correlations may reflect *holonomy in detector configuration space*, rather than explicit nonlocal causation.

This paper develops the X – Θ framework: a local, no-signaling construction in which outcomes arise from local stochastic detection coupled to a hidden phase Θ via a connection over the settings space. Version-0A is static and reproduces the familiar cosine correlator (and Tsirelson saturation at the canonical CHSH angles). Version-0B introduces bounded memory transport that makes the effective phase *path-dependent*, yielding a CW/CCW asymmetry that is not visible in static analyses.

Scope. We do not claim an experimental confirmation, nor a replacement of quantum mechanics. The aim is to propose a *testable geometric mechanism* and identify falsifiable signatures.

2 Bell scenario, CHSH, and configuration space

2.1 Operational scenario

Alice and Bob choose settings $a \in \{a_0, a_1\}$ and $b \in \{b_0, b_1\}$, and record outcomes $x, y \in \{+1, -1\}$. The correlator is

$$E(a_i, b_j) \equiv \mathbb{E}[xy \mid a = a_i, b = b_j], \quad (1)$$

and the CHSH statistic is

$$S \equiv E(a_0, b_0) + E(a_0, b_1) + E(a_1, b_0) - E(a_1, b_1). \quad (2)$$

Local realistic models satisfy $|S| \leq 2$, while quantum theory allows $|S| \leq 2\sqrt{2}$ [Tsirelson1980].

2.2 Configuration space and paths

The detector setting pairs form a discrete configuration space

$$\mathcal{C} \equiv \{(a_i, b_j)\}_{i,j \in \{0,1\}}, \quad (3)$$

a 4-vertex plaquette. A *measurement path* is an ordered sequence $\gamma = \{(a(t_k), b(t_k))\}_{k=1}^n$. Standard Bell analyses typically assume that correlators depend only on the visited vertex (a_i, b_j) , not on the order of visits. X – Θ explicitly allows *mild path dependence* via bounded internal transport, while preserving no-signaling.

3 Axioms of $X-\Theta$

We state the minimal axioms used throughout.

Axiom 1 (Hidden phase). Each emitted pair carries a shared phase $\Theta \in [0, 2\pi)$, sampled from a distribution independent of settings: $\Theta \perp (a, b)$.

Axiom 2 (Local generation and no-signaling). Outcomes x and y are generated locally from the local setting and internal variables; marginal statistics satisfy $P(x | a, b) = P(x | a)$ and $P(y | a, b) = P(y | b)$.

Axiom 3 (Nonlinear detector response). The detection map from a continuous internal signal to ± 1 outcomes is nonlinear (e.g., threshold/saturation), which acts as a harmonic filter in the phase variable and stabilizes a dominant cosine component in correlators.

Axiom 4 (Geometric coupling). Settings couple to the hidden phase via a connection over $\mathcal{C}$. In Version-0A this coupling is static. In Version-0B the effective phase transport depends on the traversal of $\mathcal{C}$ through a bounded memory variable.

Remark 1. These axioms do *not* introduce signaling or measurement dependence by construction; any such behavior would be a failure mode and must be checked in simulation or experiment.

4 Version-0A: Static connection and Tsirelson saturation

4.1 Static effective phase

In Version-0A, define the effective phase at settings (a, b) as

$$\Theta_{\text{eff}}(a, b) = \Theta + \phi(a, b), \quad (4)$$

where ϕ is a static offset induced by detector frame twists or a $U(1)$ -like connection on settings space.

Motivated by singlet correlations and geometric-phase analogues, we target correlators of the form

$$E(a, b) = -v \cos(a - b + \beta), \quad (5)$$

with visibility $v \in (0, 1]$ and a constant offset β .

4.2 CHSH and a geometric view of the plaquette

For canonical CHSH angles (e.g., $a_0 = 0$, $a_1 = \pi/4$, $b_0 = \pi/8$, $b_1 = -\pi/8$), Eq. (5) yields $|S| \approx 2\sqrt{2}v$. We interpret this as the maximal compatible “flux” of a discrete $U(1)$ connection on the CHSH plaquette, consistent with no-signaling constraints. This viewpoint echoes how gauge-invariant loop observables (Wilson loops) encode curvature [Berry1984, WilczekZee1984].

Remark 2. In this paper we emphasize operational structure rather than claiming a unique derivation of Eq. (5). A more formal derivation (e.g., from explicit connection 1-forms or discrete gauge theory on $\mathcal{C}$) is future work.

5 Version-0B: Dynamic holonomy via bounded memory transport

5.1 Bounded memory variable

Version-0B introduces a bounded internal state h (Bob-centric in the Option-B configuration) updated at each step:

$$h_{t+1} = (1 - \lambda)h_t + \delta \sigma_t, \quad (6)$$

where $\lambda \in [0, 1)$ is leakage, δ is a small geometric “kick,” and $\sigma_t \in \{+1, -1\}$ encodes orientation of the step in the settings traversal. This is not a communication channel; it is an internal phase-transport degree of freedom.

5.2 Operational holonomy and CW/CCW asymmetry

Consider traversing the CHSH plaquette $\mathcal{C}$ in a clockwise (CW) or counter-clockwise (CCW) cycle. Because σ_t changes sign under reversal, the accumulated transport differs:

$$\Omega_{\text{CW}} \neq \Omega_{\text{CCW}}. \quad (7)$$

Operationally, this implies

$$S_{\text{CW}} \neq S_{\text{CCW}}, \quad (8)$$

even though local marginals remain flat (Axiom 2).

Definition 1 (Asymmetry statistic). Define

$$\Delta S \equiv S_{\text{CW}} - S_{\text{CCW}}. \quad (9)$$

A nonzero ΔS is the primary “smoking-gun” signature of path-dependent holonomy in $\text{X-}\Theta$.

Remark 3. Standard quantum mechanics typically predicts path-independence of correlations under mere re-ordering of settings (assuming stationarity and idealized memoryless detection). $\text{X-}\Theta$ predicts a controlled violation of this path-independence while maintaining no-signaling.

6 Empirical summary from simulations

We summarize representative empirical values (from the reported configuration), emphasizing that the *relative* CW/CCW asymmetry is the central claim.

Metric	Empirical value	Physical significance
Static S (0A)	≈ -2.5998	Baseline Tsirelson-like strength (static geometry).
CW Path S (0B)	≈ -2.4794	Orientation-dependent torsion reduces strength.
CCW Path S (0B)	≈ -2.7170	Opposite orientation enhances strength.
Asymmetry ΔS	≈ 0.2375	Operational signature of chirality/holonomy.

Table 1: Representative values demonstrating path-dependent asymmetry between CW and CCW traversals of the CHSH settings loop.

Interpretation. The absolute values of S can vary with visibility, detector nonlinearity, and the effective coupling of the transported phase. The claim is that ΔS persists under controls that preserve no-signaling and measurement-independence and vanishes when the transport mechanism is disabled.

7 No-signaling and consistency checks

A central consistency requirement is that, for each fixed local setting, local outcome probabilities are independent of the remote choice. Operationally:

$$P(x = +1 \mid a_i, b_0) \approx P(x = +1 \mid a_i, b_1), \quad P(y = +1 \mid a_0, b_j) \approx P(y = +1 \mid a_1, b_j). \quad (10)$$

$\text{X-}\Theta$ enforces this by construction through symmetric local thresholding and uniform Θ sampling. Any empirical deployment must verify these equalities within statistical uncertainty, alongside confirming that ΔS is not an artifact of drift, selection bias, or time-tag asymmetries.

8 Experimental outlook

The framework suggests a direct test:

- Run a high-rate Bell test while cycling the four settings in a controlled CW order for long blocks, then CCW for long blocks.
- Compute S_{CW} , S_{CCW} , and ΔS with identical estimators and identical acceptance criteria.
- Verify no-signaling marginals in each block.
- Perform permutation or block-resampling tests to rule out time-drift explanations.

Existing photonic datasets with recorded trial order may already be suitable for retrospective analysis.

9 Future refinements (physics, not tuning)

If one seeks a larger or cleaner ΔS signal without introducing signaling or dependence on unphysical knobs, the legitimate directions are:

1. **Curved memory geometry:** let $h \in S^1$ (phase transport) rather than a scalar accumulator, making holonomy explicit.
2. **Non-Abelian transport:** promote h to a two-component state (SU(2)-like), enabling path-order sensitivity beyond mere orientation.
3. **Time-correlated switching:** incorporate dwell-time dependence $\delta \mapsto \delta(\tau)$, connecting predictions to realistic modulator dynamics.
4. **Beyond CHSH:** analyze orientation dependence in individual correlators E_{ij} , not only in the aggregate S .

These are *future work* and not required for the core claim of this paper.

10 Conclusion

We presented X- Θ , a geometric framework in which Bell-type correlations arise from a hidden phase connection over detector configuration space. Version-0A reproduces Tsirelson-level CHSH correlations as a static geometric effect, while Version-0B introduces bounded path-dependent transport that yields a CW/CCW asymmetry ΔS without enabling signaling. This suggests a falsifiable geometric signature: Bell correlations may depend weakly on the traversal orientation of settings in $\mathcal{C}$, reflecting operational holonomy rather than superluminal influence.

A Notes on interpretation and falsifiability

The X- Θ framework is falsified if:

- controlled CW/CCW cycling yields ΔS consistent with zero within tight bounds *and* systematic drift explanations are ruled out;
- any nonzero ΔS necessarily correlates with signaling marginals or other known experimental artifacts.

Conversely, a reproducible $\Delta S \neq 0$ under strict no-signaling controls would motivate deeper geometric modeling.